1 往年考题

1.1 15年考题

1. Where do software architecture come from? List five possible sources of software architecture.
2. **What distinguishes an architecture for a software product line from an architecture for a simple product?**
3. **How to model quality attribute scenarios? Graphically model two quality attributes in "stimulus-response" format: availability and performance.**
4. Describe relationships between architecture patterns and tactics. List four tactics names and describe their usage.
5. **Briefly describe the general activities involved in a software architecture process.**
6. **Mapping, and list 4 views for each style. (sa07, p.9)**
7. Explain the context, benefits and limitations of Broker Architecture Pattern.
8. Why should a software architecture be documented using different views? Give the name and purposes of 4 example views.
9. Briefly describe the fundamental principles of SOA and discuss the impact of SOA on quality attributes like interoperability, scalability and security.
10. **Describe outputs generated from each phase of ATAM process.**
11. Why SPL and MDA have high reusability? Compare and discuss their commonality and differences.

1.2 17年考题

1. **Briefly describe the general activities in a software architecture process, and the major inputs and outputs at each activity.**
2. **What distinguishes an architecture for a software product line from an architecture for a single product?**
3. What are generic design strategies applied in designing software? Give a concise working example with software architecture for each strategy.
4. **How to model quality attribute scenarios? Graphically model two quality attributes in “stimulus-response” format: availability and modifiability.**
5. **Describe outputs generated from each phase of ATAM process.**
6. **Map, and list four views of each category of style.**
7. What are ASR? List four sources and methods for extracting and identifying ASRs.
8. Please name at least three Object-Oriented principles, and explain how they are applied in Strategy pattern?
9. What should be included in a typical software architecture documentation package? Briefly describe each component and its purpose.
10. Describe 4+1 view
11. 软件设计的的三个变化维度，每个维度的变化点。differing binding time如何影响可修改性和可测试性。

Variation: forms of variation * software entity varied * binding time

1.3 18 年考题（梁神回忆）

1. 软件架构的关注点有哪些？利益相关方有哪些？
2. Software requirements, quality attributes, ASRs 的区别和联系
3. What is the nature of component-connector style? 以 MVC pattern 举例
4. 如何对质量属性场景建模？画出 availability 和 modifiability 的刺激-响应图
5. risks, sensitivity points, trade-off points 是什么？各举一个例子
6. **连线，并对每种 style 列出四种视图**
7. Layered pattern 和 Multi-tier pattern 的区别
8. 描述 ADD 过程
9. **为什么软件架构需要用不同的视图描述？举出四种视图的例子（列出名称和目的）**
10. 软件产品线架构如何实现可变性？描述可变性机制的工作方式
11. 设计一个飞行模拟软件，要求能模拟多种飞机的特性。为了在将来支持更多飞机种类，要求使用策略模式。画出架构图和类图
12. 太复杂，忘了

1.4 19年押题

这 7 个估计考的概率很大， 看明天是不是打脸吧.....

1. ATAM （看规律ATAM 和 ADD 轮流考）
2. 连线，并对每种 style 列出四种视图（每年都考，这个必然了）
3. 各种不同视图 （列举）（4+1 或者后面的）
4. 如何对质量属性场景建模？画出 availability 和 *** 的刺激-响应图 （可用性是一定要考的，其他的不好说。 概率： 性能 > 可修改性 > 其他）
5. 软件产品线.....（肯定会考，但是没法猜考什么）
6. 几种设计模式（很可能考设计题，要画图）
7. 微服务架构

1.5 19 年最终考题 （个人回忆）

1. 如何对质量属性场景建模？画出 Interoperability 和 modifiability 的刺激-响应图
2. What are ASR? List four sources and methods for extracting and identifying ASRs.
3. 4+1 视图介绍（还要画图，我的图画的的有点问题，去wiki百科上可以看）
4. What are generic design strategies applied in designing software? Give a concise working example with software architecture for each strategy. (和17年一样的)
5. Map, and list four views of each category of style.（每年必考题）
6. What should be included in a typical software architecture documentation package? Briefly describe each component and its purpose. （和17年基本一样）
7. 描述 在 ATAM 的每一个过程中 有哪些 stake holder 和他们的职责
8. 软件产品线架构如何实现可变性？描述可变性机制的工作方式，和变化点。 （和去年一样）
9. Explain the context, benefits and limitations of Broker Architecture Pattern.

10. 微服务 和 SOA 的区别，相同点
11. 一个买票系统的设计题，不同的角色有不同的打折方案，用策略模式设计， 最后画图，还要说明策略模式的使用场景。
12. 设计题，和 15 年的设计题一模一样